

MEASURE

Severity definitions (same paired landmarks, four geometries)

axis from	apex used for	test ICC
min-max line through 3 pts	axis <i>and</i> denominator	0.726
tooth-mask PCA	denominator only	0.823
CEJ → INT midpoints	denominator only	0.751
CEJ → INT midpoints	not used at all	separate scale

Evaluation

ICC(2,1) · bootstrap CI
Bland-Altman · noise study

MODELS

YOLOv8x

2-class · 200 ep · best @ 42

test mAP@0.5 **0.873**
mAP@0.5:0.95 **0.794**

infer

GT boxes

CEJ head **0.926**INT head **0.895**Apex head **0.864****Combine + pair**

6 combine modes
× 3 pairing protocols
ICC spans 0.559–0.727

paired sides

Domain shift. Heads are
trained on GT boxes, *evaluated*
on YOLO boxes.

YOLO boxes

3 × Keypoint R-CNN (ResNet-50 + FPN), 2 keypoints each

keypoint targets

LABELS

Multipass region growing

on-mask, then 1 px rings, stop at 8 px

Intersection derivation

bone line ∩ contour, ray fallback

PCA slot ordering

mesial/distal by mask long axis

v6 labels

production

retention **99.4%** CEJ / **91.0%** apex at **0.029%** cross-tooth contamination · **rejected:** v3's fixed 4 px grace lost 23.7% of apex clicks

DATA

IOPA images
1,000Tooth masks
per toothCEJ + apex clicks
6,119 / 4,490Bone-level lines
expert polylinesTooth boxes
4,402 teeth

Derived by us, not shipped by DenPAR: the single/double-root class and the bone-tooth intersection landmark. Official split 650 / 150 / 200.